10.1.1 Factors determining Chemical changes

Chemical changes will not occur at all conditions. For a chemical change to take place, certain specific conditions are required. Chemical changes can take place in the following conditions.

a. Contact in physical states
b. Solution of reactants
c. Electricity
d. Heat
e. Light
f. Catalyst

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a. Contact in physical states

We experience many events in our daily life like burning of matchstick on rubbing and iron materials turning into reddish brown. Why and how these changes happen?

These changes are due to chemical reactions by contact in physical states. Combination of reactants in their naturally occurring states (solids, liquids, gases) is referred as contact in physical states.

- When dry wood comes into contact with fire, it burns with the help of oxygen to form carbon dioxide, which is given out as smoke.
- When a matchstick is rubbed on the sides of a matchbox, a chemical reaction takes place to form heat, light and smoke.
- When quick lime (calcium oxide) comes into contact with water, it forms slaked lime (calcium hydroxide).



Figure 10.1 Burning a match stick

More to know

The head of a matchstick contains potassium chlorate and antimony trisulphide. The sides of the matchbox contain red phosphorous.

From the above reactions, we can conclude that certain chemical reactions take place only when the reactants are brought in contact with each other in their physical states.

Activity 2

Take two test tubes and couple of rust free iron nails. In one test tube pour some water and put an iron nail. Keep the test tube opened for few days. Take another test tube and pour some water along with some coconut oil. Now, place the second iron nail. Leave the set up for a few days. Observe the changes and record them. Which iron nail gets rusted and why?

b. Solution of reactants

When milk is mixed with coffee decoction the colour of the milk and the decoction changes due to chemical reaction. Similarly, when we mix two substances (reactants) in solution form, a chemical reaction takes place between them to form new substances (products). For example, take small amount of solid silver nitrate and sodium chloride in a test tube. Do you observe any change? No, because the reactants in solid state have no recation. Now, you dissolve the same reactants in water in a separate test tubes and mix both the solutions. What do you observe? Silver nitrate solution reacts with sodium chloride solution to form a white precipitate of silver chloride and sodium nitrate solution. From the above reaction, we infer that some chemical reactions proceed only in solution form not in solid form.

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